SAMADHAN AI

Progress Report

Multi-Agent Civic Complaint Management Platform

Reporting focus: completion of Agent 2 (Resolution & Routing) and the database implementation, alongside the previously completed frontend foundation and Agent 1.

1. Executive Summary

This week, the main work completed for Samadhan AI was Agent 2 — Resolution & Routing — and the database implementation. Agent 2 is responsible for taking the validated complaint analysis and preparing an appropriate resolution route, including department/jurisdiction and assignment-related decisions in accordance with the project's documented rules. The database work establishes the persistent data model needed to support complaint records and the wider complaint lifecycle.

The project also has its basic frontend and Agent 1 (Complaint Analyst) completed from earlier work. The next major work area is Agent 3 (Monitoring & Escalation), followed by integration and end-to-end validation across the platform.

2. Work Completed This Week

2.1 Agent 2 — Resolution & Routing

- Completed the second agent in the documented multi-agent workflow.
- Designed the agent's responsibility around producing routing and resolution recommendations from the validated complaint analysis supplied by Agent 1.
- Aligned routing decisions with the project's documented department/jurisdiction rules, complaint location, and assignment constraints.
- Accounted for complaints that may involve more than one department, using the documented parent-child complaint approach where applicable.
- Kept routing decisions subject to backend validation and the documented review or unassigned path when a valid destination cannot be determined.

2.2 Database Implementation

- Completed the database implementation for the project’s documented data model.
- Established the persistence foundation for complaint records and the entities needed to support routing, assignments, and complaint lifecycle tracking.
- Included the data-model concepts required by the project documentation for parent-child complaints and independent child-level tracking.
- Provided the database foundation needed for subsequent monitoring, escalation, resolution submission, citizen verification, and audit-history workflows.

3. Previously Completed Work

- Basic frontend foundation.
- Database Schema, ER Diagram
- Agent 1 — Complaint Analyst, responsible for analysing submitted complaint information and producing structured complaint analysis for downstream processing.

4. Architecture and Workflow Alignment

The work is framed around the project's documented multi-agent architecture and complaint lifecycle. The system is intended to pass structured outputs between agents rather than treating each agent as an isolated feature. Agent 2 uses the upstream complaint analysis to propose routing, while the backend and database provide the validation and persistence foundation for downstream processing.

- Composite complaints: support a parent complaint with child tickets when multiple departments are involved, with independent child assignments and SLA tracking.
- SLA handling: use deterministic, event-driven scheduling and trigger escalation processing only when an SLA breach event occurs; avoid an LLM polling loop.
- Resolution verification: an officer-submitted resolution should move into citizen verification, with the documented 48-hour response window and an independent review path for disputes.
- Agent harness: apply documented execution controls such as timeouts, token budgets, and guardrails as the agent workflow is integrated.

5. Next Steps

1. Implement Agent 3 — Monitoring & Escalation around deterministic SLA_BREACHED events rather than continuous LLM polling.
2. Integrate Agent 1, Agent 2, the backend, and database into a consistent end-to-end complaint workflow.
3. Validate parent-child complaint behaviour, independent child assignments and SLA clocks, and parent-level status synchronization.
4. Complete and validate the officer resolution → Pending_Citizen_Verification → citizen confirmation/dispute workflow, including the 48-hour window.
5. Test failure paths, uncertain routing, missing or invalid destinations, auditability, and agent execution guardrails.

6. Summary

The key deliverables for this reporting period are Agent 2 and the database implementation. Together with the existing frontend and Agent 1, these provide the core foundation for routing and persisting civic complaints. The next priority is completing Agent 3 and validating the full complaint lifecycle against the project's architecture and mentor review requirements.